



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-051 Practice Pack

Attempt-protected independent tasks using the canonical starter ZIP.

Artifact	C3-051
Status	2026 - LEARNER EDITION
Phase	C3-04 Storage, File Formats & Lakehouse 13 lessons 34-46 focused hours
Prerequisite	C3-03 Gate PASS; Bridge warehousing/lakehouse mental models assumed, implementation depth taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny example -> guided real file/table workflow -> break/fix -> independent changed task -> validation -> production reasoning -> evidence. A file extension or successful write is never sufficient proof of correct table state.



Practice rules

Each Pxx is independent work after the guided lesson. Save your actual code/config/output, one independent validation, confidence 1-4 and main uncertainty before review. If the lesson needs PyArrow/Delta runtime and the package is unavailable, save the actual failure and complete runtime evidence later - never fabricate output.



P01 - Object storage and namespace mental model

Independent task

Create architecture/01_namespace_map.md. Include landing, bronze, silver and gold locations; logical table names; owner; and one anti-pattern such as embedding a dashboard metric name inside a raw file path.

Change it so this is not copied practice

Design the namespace for a second domain (support tickets) using different physical locations but the same naming principles.

Your evidence must include

Namespace map + ownership/retention note + fresh transfer saved.

Independent validation

No generated part-file name is a business contract; raw evidence is separated from curated state; logical names remain stable.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/contract/layout. If nothing failed, inject the lesson break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control:

Do not open R01 until this attempt is saved.



P02 - CSV, JSON, Avro and Parquet trade-offs

Independent task

Complete docs/02_format_decisions.md. For each workload record read pattern, write pattern, schema needs, human readability, compression and compatibility.

Change it so this is not copied practice

Choose a format for a mobile-app event archive with nested attributes and a downstream warehouse load.

Your evidence must include

Decision matrix + changed scenario saved.

Independent validation

Decision mentions schema strength, access pattern and consumer/tool compatibility; trade-offs are explicit.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/contract/layout. If nothing failed, inject the lesson break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control:

Do not open R02 until this attempt is saved.



P03 - Columnar storage, statistics and compression

Independent task

Complete scripts/01_make_parquet_TODO.py and scripts/02_inspect_parquet_TODO.py using the current PyArrow API. Save schema, rows, row groups, selected columns and one metadata observation.

Change it so this is not copied practice

Create a fresh Parquet file from customers_day1.csv and inspect a different subset of columns.

Your evidence must include

Runnable scripts + real metadata output + parity checks.

Independent validation

Source and Parquet logical rows match; required fields/types are intentional; projected read returns only requested columns.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/contract/layout. If nothing failed, inject the lesson break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control:

Do not open R03 until this attempt is saved.



P04 - Partitioning strategy and pruning

Independent task

Complete architecture/04_partition_decision.md and generate a partition plan using src/lakehouse_core.py. State expected partition count and the query predicates the design helps.

Change it so this is not copied practice

Re-evaluate the plan for a workload that queries mostly by service_date rather than region.

Your evidence must include

Decision + partition plan + changed-query transfer.

Independent validation

Partition key is tied to actual filters; expected partition cardinality and small-file risk are documented.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/contract/layout. If nothing failed, inject the lesson break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control:

Do not open R04 until this attempt is saved.



P05 - Small-files problem and compaction

Independent task

Complete docs/05_file_health.md and run scripts/07_maintenance_audit_TODO.py after implementing the health checks. Define measurable compaction criteria.

Change it so this is not copied practice

Design a trigger for a table with fewer files but extreme size skew across regions.

Your evidence must include

Health evidence + trigger + parity plan.

Independent validation

Compaction trigger has metrics and thresholds; post-rewrite parity checks are explicit.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/contract/layout. If nothing failed, inject the lesson break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control:

Do not open R05 until this attempt is saved.



P06 - Catalogs, metadata and table namespaces

Independent task

Complete contracts/orders_current.table.json using the supplied schema. No credentials or local-machine absolute secrets belong in the contract.

Change it so this is not copied practice

Write a registration record for support.tickets_current with different ownership and retention.

Your evidence must include

Completed contract + fresh transfer.

Independent validation

Contract distinguishes logical identity from physical path and includes operational ownership.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/contract/layout. If nothing failed, inject the lesson break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control:

Do not open R06 until this attempt is saved.



P07 - Lake vs warehouse vs lakehouse + open table-format trade-offs

Independent task

Complete architecture/07_open_table_format_decision.md. Compare at least schema/partition evolution, engine/catalog support, DML/upserts, operational tooling and interoperability.

Change it so this is not copied practice

Evaluate a second scenario where the primary query engine changes from Spark to Trino.

Your evidence must include

Decision matrix + changed-engine transfer.

Independent validation

Decision is conditional, evidence-based and separates table-format semantics from storage file format.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/contract/layout. If nothing failed, inject the lesson break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control:

Do not open R07 until this attempt is saved.



P08 - Delta transaction log and ACID guarantees

Independent task

Complete scripts/03_delta_create_TODO.py. Save real runtime evidence if deltalake is installed; otherwise mark runtime pending and do not fabricate a version.

Change it so this is not copied practice

Create a fresh Delta table for customers_day1.csv at a different path.

Your evidence must include

Real version/schema/count/log evidence + fresh transfer.

Independent validation

Committed version observed; row/key parity proven; no "success" claim from file existence alone.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/contract/layout. If nothing failed, inject the lesson break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control:

Do not open R08 until this attempt is saved.



P09 - Schema enforcement and schema evolution

Independent task

Complete scripts/04_schema_evolution_TODO.py and contracts/schema_change_2026-09-02.md. Record old/new schema and allowed/blocked rationale.

Change it so this is not copied practice

Classify a fresh change that widens a numeric type and another that renames a required key.

Your evidence must include

Runtime/schema evidence + contract change + fresh classification.

Independent validation

Schema diff recorded; change approval explicit; downstream/read compatibility considered.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/contract/layout. If nothing failed, inject the lesson break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control:

Do not open R09 until this attempt is saved.



P10 - MERGE, upsert, update and delete patterns

Independent task

Complete scripts/05_merge_TODO.py using the current delta-rs TableMerger API. Save merge metrics, unique-key check and total amount before/after.

Change it so this is not copied practice

Apply a fresh correction to a different order and insert a new order with a different region.

Your evidence must include

MERGE code + metrics + independent reconciliation + rerun evidence.

Independent validation

Target has one row per order_id; expected updates/inserts occurred; rerunning the same delta is materially idempotent.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/contract/layout. If nothing failed, inject the lesson break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control: _____

Do not open R10 until this attempt is saved.



P11 - Time travel and reproducible historical reads

Independent task

Complete scripts/06_time_travel_TODO.py. Save version numbers, row/key differences and one reproducibility note.

Change it so this is not copied practice

Create another table change, then compare the two newest versions without using the earlier comparison text.

Your evidence must include

Version comparison + changed-history transfer.

Independent validation

Historical read is real runtime evidence; version difference is specific; retention implication is stated.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/contract/layout. If nothing failed, inject the lesson break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control:

Do not open R11 until this attempt is saved.



P12 - Medallion/layered architecture and table contracts

Independent task

Complete contracts/bronze.json, silver.json and gold.json. Each must include purpose, grain, required fields, owner, quality, refresh/replay and allowed change policy.

Change it so this is not copied practice

Design layers for support events where gold is a daily SLA dashboard aggregate.

Your evidence must include

Three contracts + fresh domain transfer.

Independent validation

Every layer has a distinct purpose/grain; no source evidence is destroyed; consumer dependency is explicit.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/contract/layout. If nothing failed, inject the lesson break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control:

Do not open R12 until this attempt is saved.



P13 - Optimization, retention and maintenance strategy

Independent task

Complete runbooks/MAINTENANCE_TODO.md and scripts/07_maintenance_audit_TODO.py. Never run destructive vacuum merely to complete the lesson.

Change it so this is not copied practice

Design a policy for a regulated table requiring 90-day reproducibility and compare it with a low-risk append-only staging table.

Your evidence must include

Runbook + dry-run design + changed-policy transfer.

Independent validation

No destructive cleanup occurs without recovery-window check; maintenance has before/after integrity controls.

Failure/recovery note

Record one failure, surprise or uncertainty. State what evidence you inspected before changing code/contract/layout. If nothing failed, inject the lesson break/fix case and recover it.

Attempt log - append only

Attempt ID/date: _____ | confidence 1-4: ____ | main uncertainty: _____ | independent control:

Do not open R13 until this attempt is saved.